

WONTHAGGI, Australia

WMO: 958810

Lat: 38.608S

Lon: 145.596E

Elev: 42

StdP: 100.82

Time Zone: 10.00 (AUS)

Period: 90-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 4.0	(c) 5.1	(d) 2.0	(e) 4.4	(f) 7.8	(g) 2.8	(h) 4.6	(i) 8.0	(j) 18.4	(k) 12.5	(l) 14.9	(m) 11.2	(n) 0.4	(o) 270	(p) 0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 9.3	(c) 32.0	(d) 20.6	(e) 29.0	(f) 20.1	(g) 26.3	(h) 19.0	(i) 21.8	(j) 28.6	(k) 20.7	(l) 26.9	(m) 19.7	(n) 25.4	(o) 4.8	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.9	(b) 14.6	(c) 24.7	(d) 18.6	(e) 13.5	(f) 23.2	(g) 17.6	(h) 12.7	(i) 22.2	(j) 63.8	(k) 28.7	(l) 59.9	(m) 26.8	(n) 56.4	(o) 25.4	(p) 34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.8	12.1	10.0	1.3	37.4	1.2	3.5	0.4	39.9	-0.2	42.0	-0.9	43.9	-1.7	46.4	
			1.0	24.6	1.0	1.5	0.3	25.7	-0.3	26.6	-0.8	27.4	-1.6	28.5	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	18.4	19.3	17.0	14.6	12.5	10.9	10.0	10.5	11.4	13.2	14.6	16.4
	DBStd	3.98	3.25	3.74	2.80	2.57	1.93	1.79	1.65	1.85	2.26	2.59	2.80	3.20
	HDD10.0	54	0	0	0	1	2	11	19	14	7	1	1	0
	HDD18.3	1716	37	28	60	117	181	224	257	243	207	161	119	81
	CDD10.0	1529	262	259	218	138	79	37	21	30	50	101	137	197
	CDD18.3	149	41	53	21	5	0	0	0	0	1	3	6	20
	CDH23.3	1336	348	471	173	31	0	0	0	0	2	36	69	207
	CDH26.7	527	147	202	59	4	0	0	0	0	0	6	23	85
Wind	WSAvg	3.3	3.7	3.2	3.0	2.6	2.6	3.0	3.1	3.6	3.9	3.4	3.6	3.8
Precipitation	PrecAvg	954	53	42	59	77	102	96	108	105	94	90	70	60
	PrecMax	1379	150	187	191	191	274	175	209	216	219	165	138	143
	PrecMin	619	4	0	3	19	33	38	33	27	18	28	16	8
	PrecStd	171	29	40	34	42	48	38	48	41	40	37	32	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	36.0	32.2	26.3	21.7	18.1	16.4	19.5	23.0	27.0	30.1	33.6
		MCWB	20.7	21.1	20.1	16.7	15.2	13.8	11.8	13.9	16.2	17.9	19.8	21.4
	2%	DB	30.8	32.0	27.7	23.3	19.2	16.0	14.9	16.9	19.9	23.5	25.0	29.5
		MCWB	20.6	20.8	19.3	16.5	14.4	13.0	11.2	12.3	14.4	16.5	18.3	19.2
	5%	DB	27.2	28.7	24.9	21.1	17.5	14.9	13.8	15.4	17.7	20.5	22.0	25.4
		MCWB	20.0	20.4	18.3	15.9	13.9	12.2	10.8	11.6	13.2	15.1	17.1	18.4
	10%	DB	24.2	25.6	22.1	18.9	16.2	13.9	12.9	14.1	15.8	18.0	19.5	22.2
		MCWB	18.9	19.5	17.6	15.1	13.3	11.5	10.3	10.8	12.4	14.2	15.8	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	24.4	21.5	19.3	16.6	14.6	12.9	15.1	17.0	19.0	21.0	22.1
		MCDB	29.6	32.5	28.8	23.5	19.6	17.1	14.7	18.7	20.9	25.0	26.4	29.8
	2%	WB	21.5	22.1	19.9	17.5	15.2	13.6	12.1	13.1	15.5	17.0	19.1	20.1
		MCDB	28.7	28.2	25.3	21.6	17.7	15.3	13.9	16.0	19.5	21.9	24.6	27.4
	5%	WB	20.3	20.9	18.7	16.3	14.4	12.6	11.3	12.0	13.8	15.7	17.4	18.9
		MCDB	26.2	26.6	24.0	19.8	16.8	14.3	13.1	14.7	16.5	19.9	21.4	24.5
	10%	WB	19.1	19.8	17.6	15.4	13.6	11.8	10.7	11.1	12.7	14.4	16.0	17.6
		MCDB	23.9	25.5	21.7	18.3	15.9	13.6	12.4	13.6	15.3	17.7	19.4	22.1
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.3	7.9	7.7	6.2	5.3	5.6	6.2	6.7	7.5	7.9	8.2
		MCDBR	15.2	15.5	13.9	12.2	8.7	7.3	6.9	8.9	10.6	13.0	13.5	16.0
		MCWBR	7.3	7.3	6.7	6.3	5.5	4.9	4.5	5.5	6.7	7.3	8.3	8.1
	5% WB	MCDBR	13.4	12.8	12.2	10.3	7.8	6.0	6.5	7.9	9.3	11.9	12.5	12.8
		MCWBR	7.3	7.2	6.6	6.0	5.5	4.5	4.7	5.4	6.4	7.4	8.1	7.5
Clear-Sky Solar Irradiance	taub	0.358	0.358	0.339	0.332	0.316	0.309	0.307	0.320	0.345	0.343	0.349	0.345	
	taud	2.490	2.501	2.553	2.550	2.577	2.580	2.581	2.539	2.462	2.484	2.492	2.522	
	Ebn at Noon	973	948	927	868	823	797	824	869	904	953	976	992	
	Edh at Noon	114	108	94	82	69	64	68	82	101	108	113	111	
All-Sky Solar Radiation	RadAvg	7.01	6.11	4.81	3.34	2.20	1.76	1.98	2.81	4.02	5.39	6.34	7.04	
	RadStd	0.43	0.31	0.26	0.17	0.11	0.12	0.08	0.14	0.24	0.36	0.34	0.39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.50	N/A	N/A	+0.79	N/A	N/A	-16	-132	+173	+57

Nomenclature: See separate page